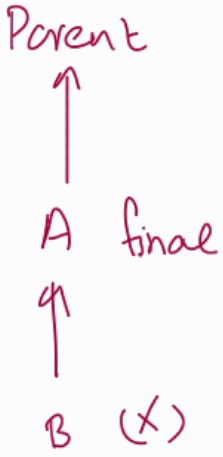
1) Final class is a class that cannot be inherited.

This means no other class can be a subclass of a final class




```
class Last final {
  public:
    void display() {
      cout<< "this is final";
    }
};

class Subclass : public Last
{
    error;
};
```

2) concrete class (Normal class)

that we can create objects from

```
class Car {
  public:
    void start() { cout<< "hello" };

Car C;
```

3) Abstract class → a class that has atleast 1 pure virtual function.

You cannot create objects from this class.

```
class Shape {
  public:
    virtual void draw() = 0;
```

};

Shape S; (X)

4) Polymorphic class → a class that has atleast one virtual function.

```
class Animal {
    public:
        virtual void Speak() { cout << "hello guys"; }
};
```

can create objects, override its virtual function with derived classes.

Pure virtual → is a function in a class that has no Body, only declaration.

virtual void Speak() = 0;

↓

means this function has no implementation here, derived classes must override it

```
class Animal {
    public:
        virtual void Speak() = 0;
};
```

Animal doesn't know how to Speak.
every specific animal must define it

```
class Dog : public Animal {
    public
    void Speak() {
        cout << "woof";
```

3

3;

Animal a (X)

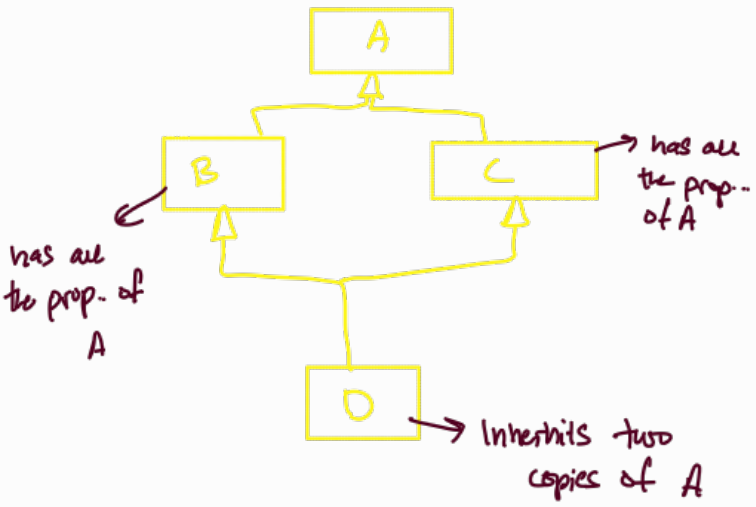
Dog a (✓) → we can create the object for Dog as it gives us the complete code for Speak() and is not abstract.

Virtual Base class (used in Inheritance diamond problem).

↓

used to avoid duplicate copies of a base class when using multiple inheritence

```
class A {};
class B : public A {};
class C : pobic A {};
class D: pubic B, public C{};.
```



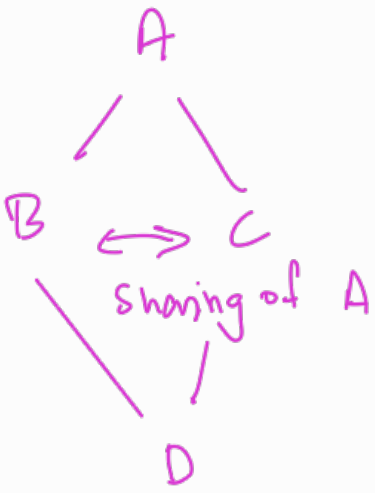
D must Inherit only 1 copy of A and not 2.

This causes confusion for the complier → which A class should D uses its constructors?

Here to solve this we Make A a virtual Base class.

B, C Inherits virtually

```
class A : {}
class B : virtual public A {};
class C : virtual public A {};
class D: pubic B, public C{};. → D Inherits only a Single A as Both B and C share 1 A.
```



B and C inherit A virtually meaning they have only 1 copy of A.